11.5.1 Electronic configuration of atoms

You already know that electrons occupy different energy levels called orbits or shells. The distribution of electrons in different shells is called electronic configuration. This distribution of electrons is governed by certain rules or conditions, known as Bohr and Bury Rules of electronic configuration.

Rule 1: The maximum number of electrons that can be accommodated in a shell is equal to $2n^2$ where 'n' is the serial number of the shell from the nucleus.

Shell	Value of (n)	Maximum number of electrons ($2n^2$)
K	1	$2 \times 1^2 = 2$
L	2	$2 \times 2^2 = 8$
M	3	$2 \times 3^2 = 18$
N	4	$2 \times 4^2 = 32$

Rule 2: Shells are filled in a **stepwise manner** in the increasing order of energy.

Rule 3: The outermost shell of an atom cannot have more than 8 electrons , even if it has capacity to accomodate more electrons. For example, electronic arrangement in calcium having 20 electrons is,

K	L	M	N
2	8	8	2

Problem 2

What is the Electronic configuration of Aluminium?

Solution

Electronic configuration of Aluminium atom: (Z = 13) **K shell = 2 , L shell = 8 and M shell = 3 electron**.

So its electronic configuration is 2, 8, 3

DO YOU KNOW?

The forces between the protons and the neutrons in the nucleus are of special kind called Yukawa forces. This strong force is more powerful than gravity.

Geometric Representation of atomic structure of elements

Knowing the mass number and atomic number of an element we can represent atomic structure.

Example:

Geometric Representation of oxygen atom $^{16}_{8}O$

Mass number A = 16

Atomic number Z = 8

Number of neutrons = A – Z = 16 – 8 = 8

Number of protons = 8

Number of electron = 8

Electronic configuration = 2, 6

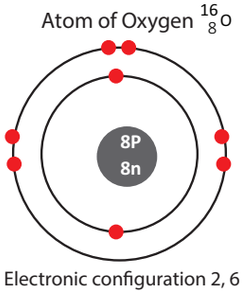


Figure 11.7 Atom of oxygen.

DO YOU KNOW?

Atoms are so tiny their mass number cannot be expressed in grams but expressed in amu (atomic mass unit). New unit is U. Size of an atom can be measured in nano metre (1 nm = 10^{-9} m) Even though atom is an invisible tiny particle now-a-days atoms can be viewed through SEM that is Scanning Electron Microscope.

11.5.2 Valence electrons

In the above example, we can see that there are six electrons in the outermost shell of oxygen atom. These six electrons are called as valence electrons.

The outermost shell of an atom is called valence shell and the electrons present in the valence shell are known as valence electrons. The chemical properties of elements are decided by these valence electrons, since they are the ones that take part in chemical reactions.

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